

Jindong Cao

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Skills

Languages: Python, C++, Java, SQL, TypeScript, Bash

Machine Learning Frameworks: PyTorch, TensorFlow

Machine Learning: Recommendation Systems, LLM, VLM, Diffusion Model, Multi-Task Learning

Tools: Git, Linux, Docker, vLLM, Spark

Education

Carnegie Mellon University, College of Engineering

M.S. in Electrical and Computer Engineering, GPA: 3.75/4.0

Pittsburgh, USA

Aug 2023 – Dec 2024

Sun Yat-sen University, School of Computer Science and Engineering

B.S. in Computer Science and Technology, GPA: 3.9/4.0

Guangzhou, China

Aug 2019 – Jul 2023

Experience

TikTok

San Jose, USA

Machine Learning Engineer II

Jan 2025 – Present

Languages: Python, C++, SQL; Skills: TensorFlow, Recommendation Systems, Hive, LLM/VLM

- Promoted from **MLE I to MLE II** within 12 months.
- Owned the end-to-end **large-scale push ranking system**, spanning Hive/SQL data pipelines, feature engineering, sample and label construction, deep ranking model design, offline evaluation, and online serving and experimentation.
- Redesigned the video push ranking model from single-objective CTR prediction to multi-task learning with Shared-Bottom and Multi-Tower architectures, jointly optimizing Click, Like, and Finish objectives to mitigate clickbait and improve DAU among mid-low active users by **+0.22%**.
- Designed E-Commerce push candidate retrieval, ranking models, and features, delivering **+0.61%** per-capita GMV and **+0.34%** paid active days.
- Built a unified multi-business MixRank framework in C++, designing multi-objective scoring functions to rank candidates across business scenarios and balance engagement and business-specific goals, delivering **+0.18% DAU**.
- Built a generative recommendation system using LLMs/VLMs, conditioning on user interest profiles and historical content consumption to generate high-interest push candidates, improving LLM-as-Judge relevance by **140%**.

Amazon

San Diego, USA

Software Development Engineer Intern

May 2024 – Aug 2024

Languages: React, Java, TypeScript, SQL

- Built an internal payment investigation platform with React, Java, and TypeScript, reducing manual payment-record lookup workload by 60% and improving operational efficiency.

ZMO.AI

Shenzhen, China

Machine Learning Engineer Intern

Mar 2023 – May 2023

Languages: Python; Skills: PyTorch, LLM, Diffusion Model and Multi-Modal Learning

- Built an image editing pipeline controlled through natural language, integrating ChatGPT, SAM, GroundingDINO, and Stable Diffusion.
- Applied pruning to BLIP-2 and CLIP, reducing memory usage by 30% and runtime by 40%.

Huawei Technologies

Dongguan, China

Software Development Engineer Intern

Jul 2022 – Sep 2022

Languages: C++; Skills: Data Structure

- Refactored a data center query system in C++, reducing average query latency by 70%.

Research

Pattern Recognition

Pittsburgh, US

Co-First Author, Weakly Supervised Prompt Learning for Low-Resource Medical Image Classification Jan 2024

- Developed **MedPrompt**, a weakly supervised prompt learning framework that combines vision-language pretraining with automatic prompt generation for low-resource medical image classification.
- Outperformed existing prompt learning and transfer learning methods across multiple medical imaging datasets, achieving up to **10.4%** improvement in few-shot classification.